

ENME608: Mechanism and Dynamics of Machinery

Lecture 4: Linkage Position Analysis (Part 2)

Dr. Thein Than Tun

Lecturer – Mechatronics & Dynamics

School of Engineering, Computer and Mathematical Sciences



Assessment

Assessment Type	Weighting	Due Date
Mid-semester Test	25%	1 st April 2026 7pm (1 hour duration) Check Venue on Canvas
Assignment	25%	31 May 2026, Online Submission by 11:59 PM
Final Controlled Assessment	50%	After Week 12 (Date & Time: TBA)

Contents of ENME608

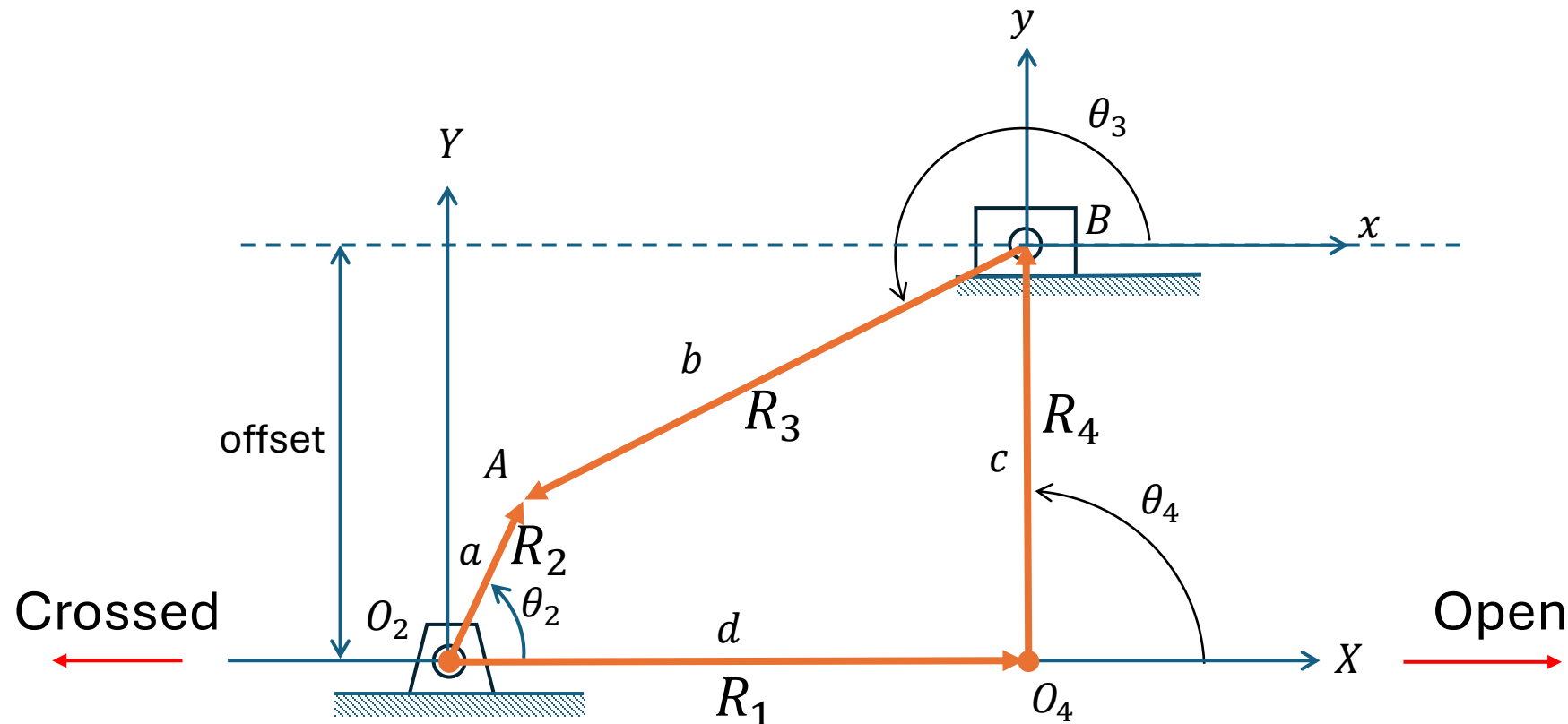
Mid-Sem Break	Mechanism and Machine Concepts	Week 1	Contents of Mid-Semester Test
	Linkages	Week 2	
	Position Analysis	Week 3 – 4	
	Velocity Analysis	Week 5	
	Acceleration Analysis	Week 6	
	Cam	Week 7	Contents of Final Exam
	Cam (continued) and Gear	Week 8	
	Gear (continued)	Week 9	
	Dynamics Fundamentals	Week 10	
	Force Analysis	Week 11	
	Force Analysis (continued) and revision	Week 12	
Final Exam (TBA)			

Reference

- Recommended Textbook
 - Robert L. Norton, *Design of Machinery : An Introduction to the Synthesis and Analysis of Mechanisms and Machines*, 6th Edition, McGraw Hill, 2019, ISBN-10: 1260113310 , ISBN-13: 978-1260113310
(Online version available: AUT Library or
ENME608: Course Resources in Canvas)

Position Analysis of Crank-Slider: Vector Loop

$$\text{Vector Loop: } R_2 - R_3 - R_4 - R_1 = 0$$



Vectors	Link Length
R_2	a
R_3	b
R_4	c
R_1	d

Position Analysis of Crank-Slider: Vector Loop (Cont.)

$$R_2 - R_3 - R_4 - R_1 = 0$$

$$ae^{j\theta_2} - be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} = 0$$

$\theta_4 = \underline{\hspace{2cm}}$
 $\theta_1 = \underline{\hspace{2cm}}$

$$a(\cos \theta_2 + j \sin \theta_2) - b(\cos \theta_3 + j \sin \theta_3) - jc - d = 0$$

$$\text{Real : } a \cos \theta_2 - b \cos \theta_3 - d = 0$$

$$\text{Imaginary : } a \sin \theta_2 - b \sin \theta_3 - c = 0$$

} $\underline{\hspace{2cm}}$ **Eqns.**
 $\underline{\hspace{2cm}}$ **Unknown**

$$\theta_3 = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right), \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi$$

$$d = a \cos \theta_2 - b \cos \theta_3$$

Position Analysis of Crank-Slider: Example

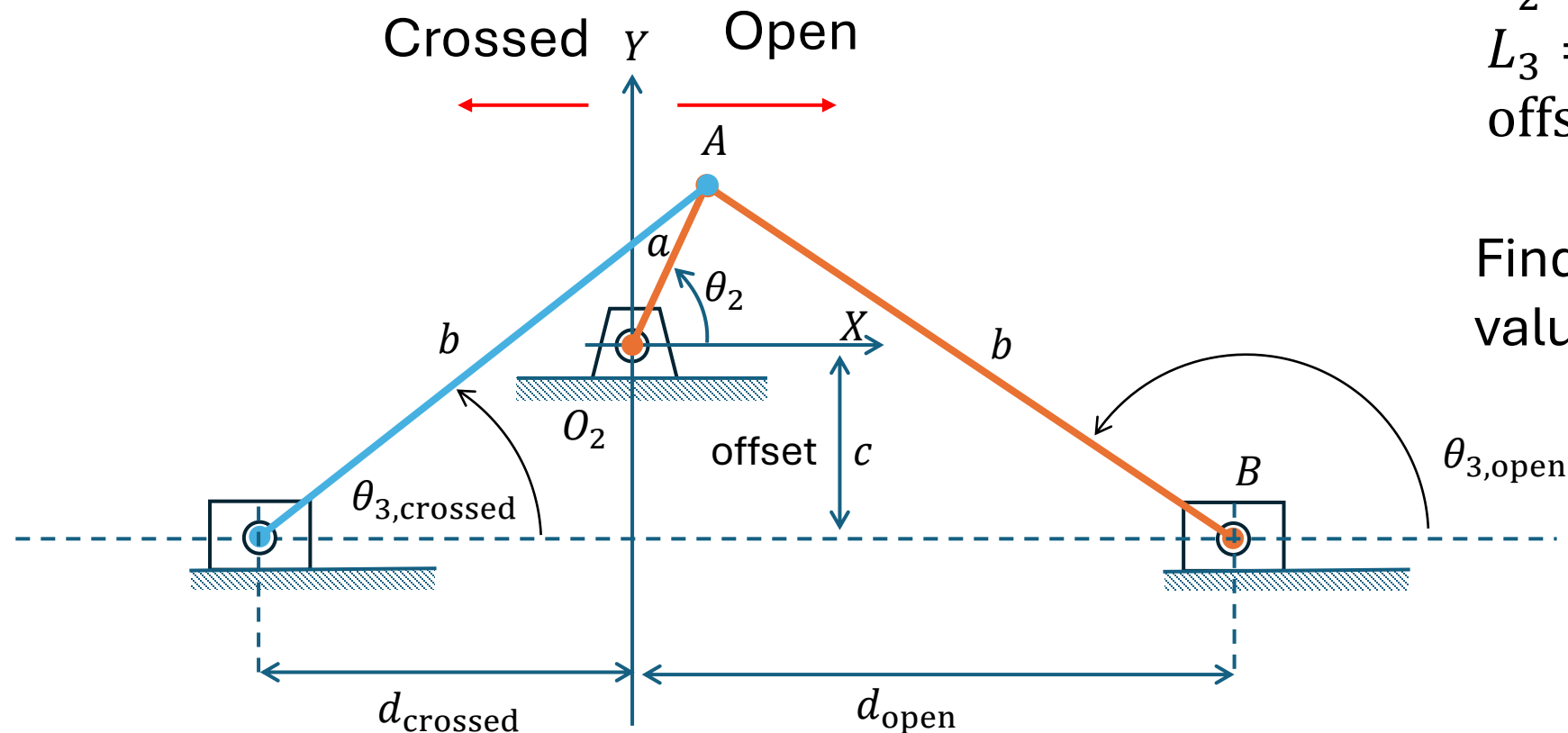
Given a fourbar linkage with the link lengths

$$L_2 = a = 40\text{mm}$$

$$L_3 = b = 120\text{mm}$$

$$\text{offset} = c = -20\text{mm},$$

Find $\theta_2 = 60^\circ$, find all possible values of θ_3 and d .



Position Analysis of Crank-Slider: Example

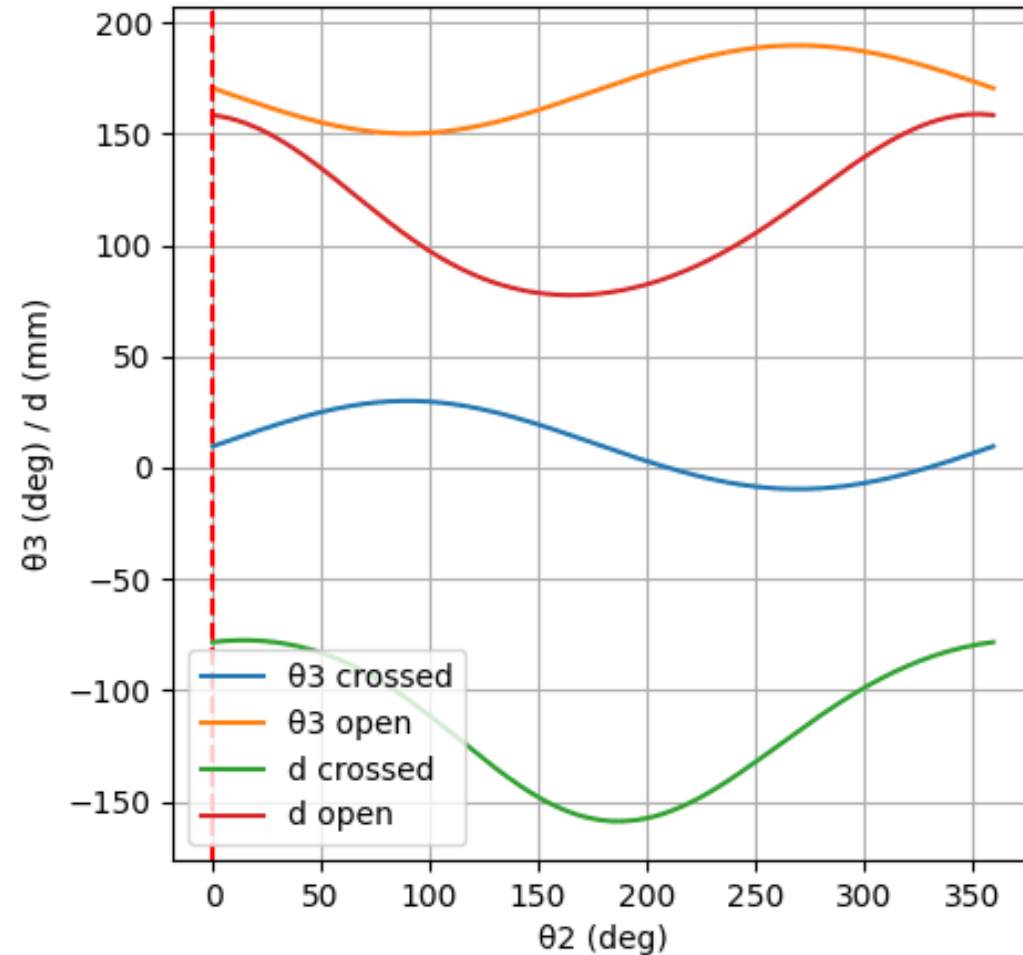
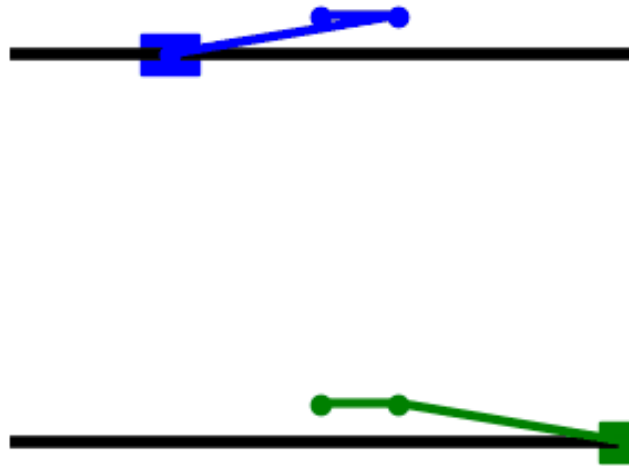
$$\theta_{3_{open}} = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right) = \arcsin\left(\frac{40 \sin(60^\circ) - (-20)}{120}\right) = 152.91^\circ$$

$$d = a \cos \theta_2 - b \cos \theta_3 = 40 \cos(60^\circ) - 120 \cos(152.91^\circ) = 126.84 \text{ mm}$$

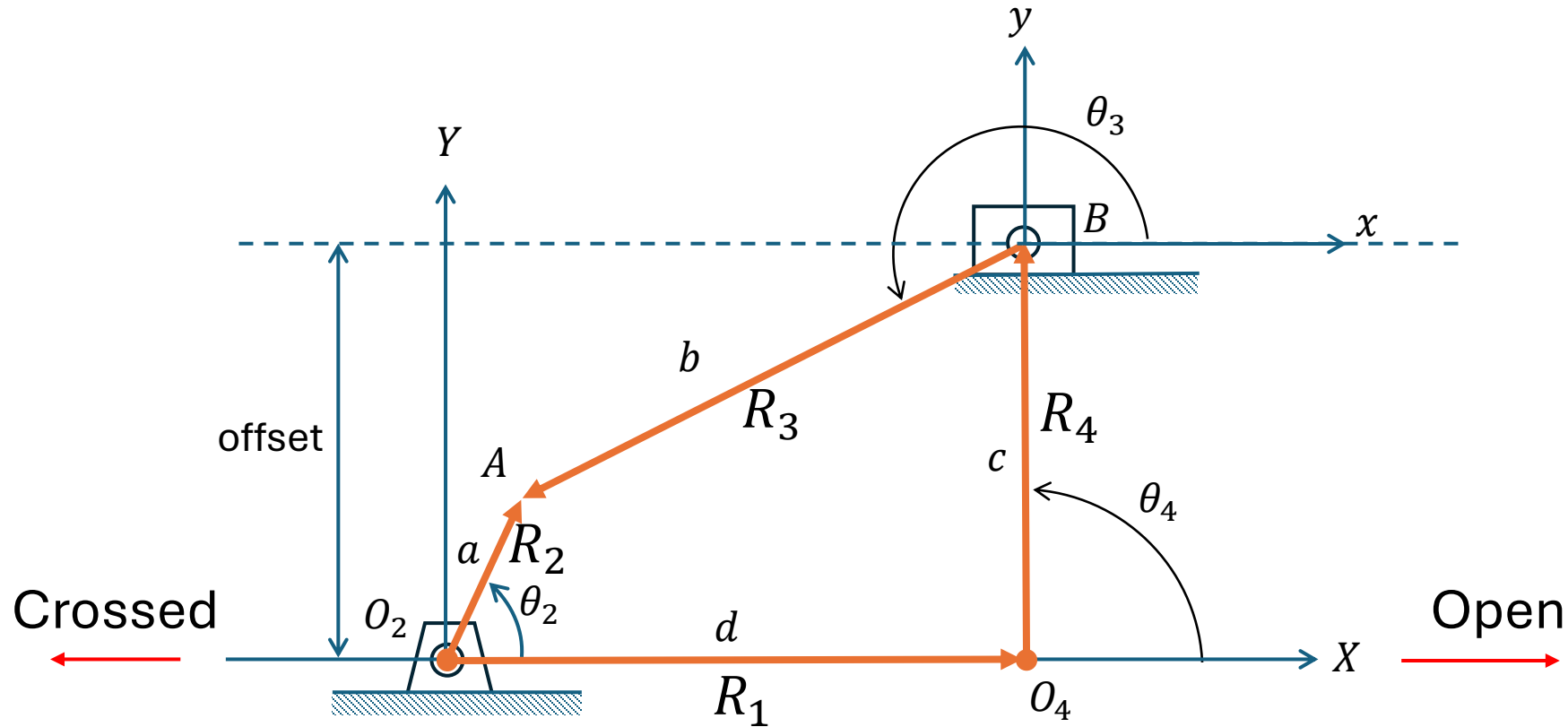
$$\theta_{3_{crossed}} = \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi = \arcsin\left(-\frac{40 \sin(60^\circ) - (-20)}{120}\right) + \pi = 27.09^\circ$$

$$d = a \cos \theta_2 - b \cos \theta_3 = 40 \cos(60^\circ) - 120 \cos(27.09^\circ) = -86.84 \text{ mm}$$

Position Analysis of Crank-Slider: Example (Cont.)



Position Analysis of Slider-Crank: Algebraic



$$\begin{array}{lcl}
 \text{Real : } a \cos \theta_2 - b \cos \theta_3 - d = 0 & \left. \begin{array}{l} \text{--- Eqns.} \\ \text{--- Unknown} \end{array} \right\} & \\
 \text{Imaginary : } a \sin \theta_2 - b \sin \theta_3 - c = 0 & &
 \end{array}$$

Position Analysis of Slider-Crank: Algebraic (Cont.)

$$\begin{array}{lcl} a \cos \theta_2 - b \cos \theta_3 - d = 0 & \Leftrightarrow & b \cos \theta_3 = a \cos \theta_2 - d \\ a \sin \theta_2 - b \sin \theta_3 - c = 0 & \Leftrightarrow & b \sin \theta_3 = a \sin \theta_2 - c \end{array}$$

$$K_1 + K_2 \sin \theta_2 + K_3 \cos \theta_2 = 0$$

where $K_1 = a^2 - b^2 + c^2 + d^2,$
 $K_2 = -2ac,$
 $K_3 = -2ad$

Position Analysis of Slider-Crank: Algebraic (Cont.)

Recall half-angle identities

$$\sin\theta_2 = \frac{2\tan\left(\frac{\theta_2}{2}\right)}{1 + \tan^2\left(\frac{\theta_2}{2}\right)}, \quad \cos\theta_2 = \frac{1 - \tan^2\left(\frac{\theta_2}{2}\right)}{1 + \tan^2\left(\frac{\theta_2}{2}\right)} \rightarrow K_1 + K_2 \sin\theta_2 + K_3 \cos\theta_2 = 0$$

$$K_1 + K_2 \left(\frac{2\tan\left(\frac{\theta_2}{2}\right)}{1 + \tan^2\left(\frac{\theta_2}{2}\right)} \right) + K_3 \left(\frac{1 - \tan^2\left(\frac{\theta_2}{2}\right)}{1 + \tan^2\left(\frac{\theta_2}{2}\right)} \right) = 0$$

$$\theta_{2_{1,2}} = 2 \arctan \left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \right)$$

where

$$\begin{aligned} A &= K_1 - K_3, \\ B &= 2K_2, \\ C &= K_1 + K_3 \end{aligned}$$

Position Analysis of Slider-Crank: Algebraic (Cont.)

$$\theta_{2_{1,2}} = 2 \arctan \left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \right)$$

where $A = K_1 - K_3,$
 $B = 2K_2,$
 $C = K_1 + K_3$

$$\theta_{3_1} = \arcsin \left(\frac{a \sin \theta_2 - c}{b} \right) \text{ or } \theta_{3_2} = \arcsin \left(-\frac{a \sin \theta_2 - c}{b} \right) + \pi$$

Using (θ_2, θ_3) pair,

Check $d = a \cos \theta_2 - b \cos \theta_3$ ***

Position Analysis of Slider-Crank: Example

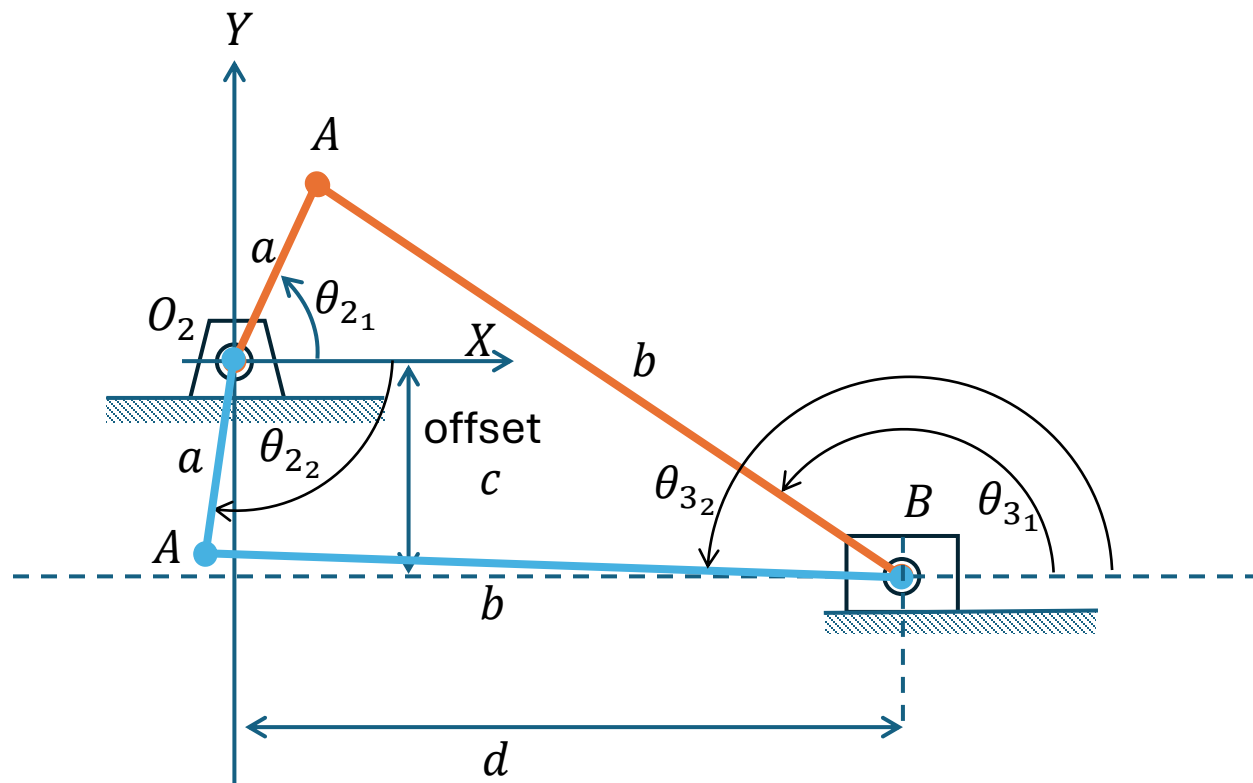
$$\theta_{3_1} = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right), \theta_{3_2} = \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi$$
$$d = a \cos \theta_2 - b \cos \theta_3$$

	θ_{2_1}	θ_{2_2}
θ_{3_1}	$d_{11} = a \cos \theta_{2_1} - b \cos \theta_{3_1}$	$d_{21} = a \cos \theta_{2_2} - b \cos \theta_{3_1}$
θ_{3_2}	$d_{12} = a \cos \theta_{2_1} - b \cos \theta_{3_2}$	$d_{22} = a \cos \theta_{2_2} - b \cos \theta_{3_2}$

Identify θ_{3_1} or θ_{3_2} formula which produces the valid d value.

Calculate $\theta_{3_1}^*$ and $\theta_{3_2}^*$ using the identified θ_{3_1} or θ_{3_2} with θ_{2_1} and θ_{2_2} .

Position Analysis of Slider-Crank: Example



Given a fourbar linkage with the link lengths

$$L_2 = a = 40\text{mm}$$

$$L_3 = b = 120\text{mm}$$

$$\text{offset} = c = -20\text{mm},$$

Find $d = 100\text{mm}$, find all possible values of θ_2 and θ_3 .

Position Analysis of Slider-Crank: Example

$$K_1 = a^2 - b^2 + c^2 + d^2 = 40^2 - 120^2 + (-20)^2 + 100^2 = -2400$$

$$K_2 = -2ac = -2(40)(-20) = 1600$$

$$K_3 = -2ad = -2(40)(100) = -8000$$

$$A = K_1 - K_3 = -2400 - (-8000) = 5600$$

$$B = 2K_2 = 2(1600) = 3200$$

$$C = K_1 + K_3 = -2400 + (-8000) = -10400$$

$$\theta_{2_1} = 2 \tan^{-1} \left(\frac{-B + \sqrt{B^2 - 4AC}}{2A} \right) = 2 \tan^{-1} \left(\frac{-3200 + \sqrt{3200^2 - 4(5600)(-10400)}}{2(5600)} \right) = 95.798^\circ$$

$$\theta_{2_2} = 2 \tan^{-1} \left(\frac{-B - \sqrt{B^2 - 4AC}}{2A} \right) = 2 \tan^{-1} \left(\frac{-3200 - \sqrt{3200^2 - 4(5600)(-10400)}}{2(5600)} \right) = -118.418^\circ$$

Position Analysis of Slider-Crank: Example

$$\theta_{3_1} = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right), \theta_{3_2} = \arcsin\left(-\frac{a \sin \theta_2 - c}{b}\right) + \pi$$

$$d = a \cos \theta_2 - b \cos \theta_3$$

	θ_{2_1}	θ_{2_2}
θ_{3_1}	$d_{11} = a \cos \theta_{2_1} - b \cos \theta_{3_1}$	$d_{21} = a \cos \theta_{2_2} - b \cos \theta_{3_1}$
θ_{3_2}	$d_{12} = a \cos \theta_{2_1} - b \cos \theta_{3_2}$	$d_{22} = a \cos \theta_{2_2} - b \cos \theta_{3_2}$

Using (θ_2, θ_3) pair,

Check $d = a \cos \theta_2 - b \cos \theta_3$ ***

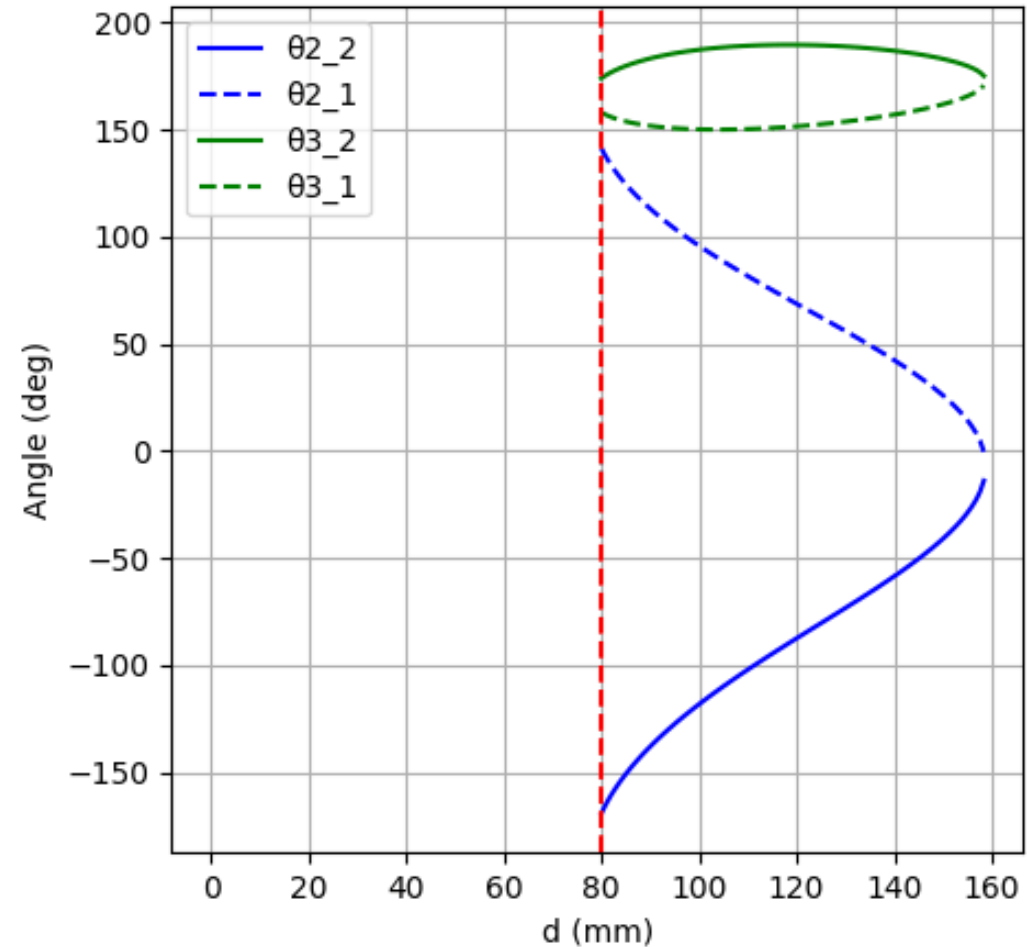
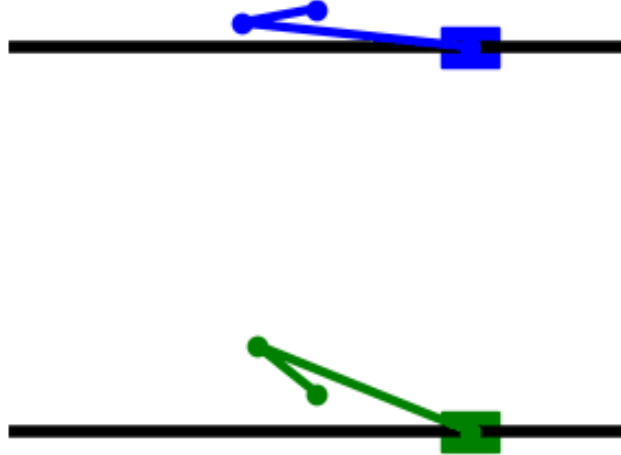
Position Analysis of Slider-Crank: Example

	θ_{2_1}	θ_{2_2}
θ_{3_1}	$d_{11} = a \cos \theta_{2_1} - b \cos \theta_{3_1}$	$d_{21} = a \cos \theta_{2_2} - b \cos \theta_{3_1}$
θ_{3_2}	$d_{12} = a \cos \theta_{2_1} - b \cos \theta_{3_2}$	$d_{22} = a \cos \theta_{2_2} - b \cos \theta_{3_2}$

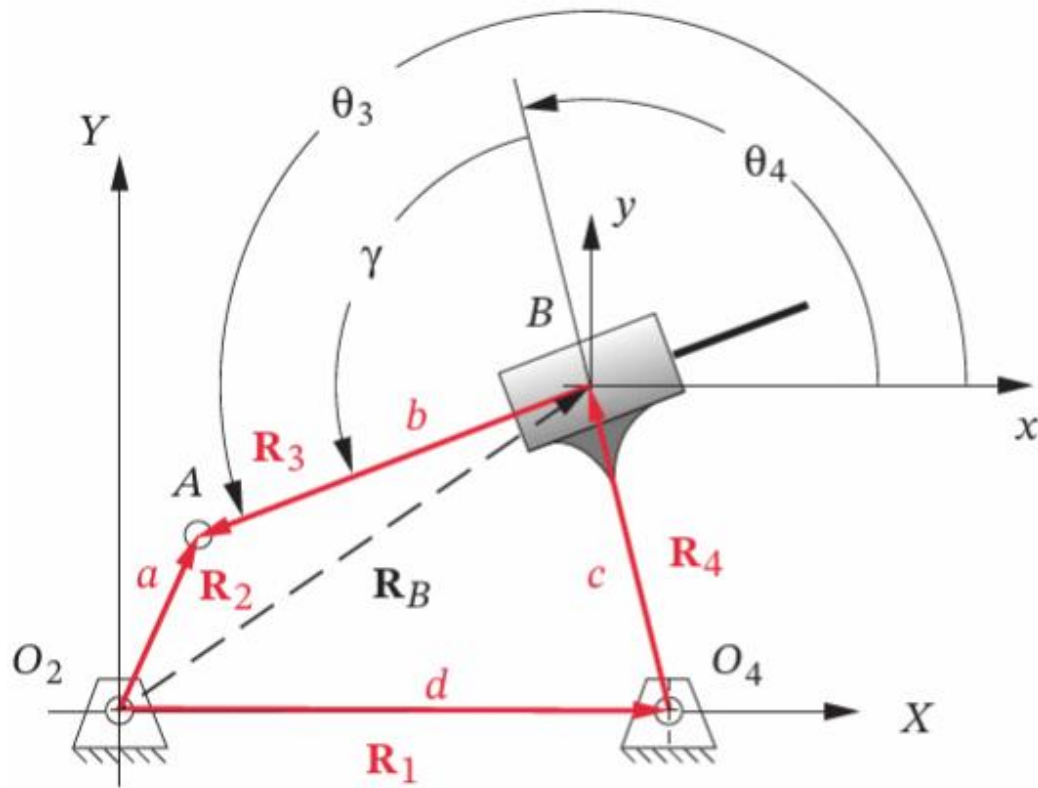
$$\theta_{3_1}^* = \arcsin \left(-\frac{a \sin(\overbrace{95.798}^{\theta_{2_1}}) - c}{b} \right) + \pi = 150.113^\circ$$

$$\theta_{3_2}^* = \arcsin \left(-\frac{a \sin(\overbrace{-118.418}^{\theta_{2_2}}) - c}{b} \right) + \pi = 187.267^\circ$$

Position Analysis of Slider-Crank: Example (Cont.)



Position Analysis of Inverted Crank-Slider: Vector Loop



$$\text{Vector Loop: } R_2 - R_3 - R_4 - R_1 = 0$$

Position Analysis of Inverted Crank-Slider: Vector Loop (Cont.)

$$R_2 - R_3 - R_4 - R_1 = 0$$

$$ae^{j\theta_2} - be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} = 0$$

$\theta_4 = \rule{1cm}{0.4pt}$ $\theta_1 = \rule{1cm}{0.4pt}$

$$a(\cos \theta_2 + j \sin \theta_2) - b(\cos \theta_3 + j \sin \theta_3) - c(\cos \theta_4 + j \sin \theta_4) - d = 0$$

Imaginary : $a \sin \theta_2 - b \sin \theta_3 - c \sin \theta_4 = 0 \quad \Leftrightarrow \quad b = \frac{a \sin \theta_2 - c \sin \theta_4}{\sin \theta_3}$

Real : $a \cos \theta_2 - b \cos \theta_3 - c \cos \theta_4 - d = 0$

$$a \cos \theta_2 - \frac{a \sin \theta_2 - c \sin \theta_4}{\sin \theta_3} \cos \theta_3 - c \cos \theta_4 - d = 0$$

Position Analysis of Inverted Crank-Slider: Vector Loop (Cont.)

Relationship between θ_3 and θ_4 is

open configuration: $\theta_3 = \theta_4 + \gamma$; crossed configuration: $\theta_3 = \theta_4 + \gamma - \pi$

Substitution

$$a \cos \theta_2 - \frac{a \sin \theta_2 - c \sin \theta_4}{\sin \theta_3} \cos \theta_3 - c \cos \theta_4 - d = 0$$

Result

$$P \sin \theta_4 + Q \cos \theta_4 + R = 0$$

where

$$P = a \sin \theta_2 \sin \gamma + (a \cos \theta_2 - d) \cos \gamma$$

$$Q = -a \sin \theta_2 \cos \gamma + (a \cos \theta_2 - d) \sin \gamma$$

$$R = -c \sin \gamma$$

Position Analysis of Inverted Crank-Slider: Vector Loop (Cont.)

Recall half-angle identities

$$\sin\theta_4 = \frac{2\tan\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)}, \quad \cos\theta_4 = \frac{1 - \tan^2\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)}$$

$$\theta_{4_{1,2}} = 2 \arctan\left(\frac{-T \pm \sqrt{T^2 - 4SU}}{2S}\right)$$

$\Leftrightarrow$

$$S \tan^2\left(\frac{\theta_4}{2}\right) + T \tan\left(\frac{\theta_4}{2}\right) + U = 0$$

where $S = R - Q$, $T = 2P$, $U = Q + R$

$$P \sin\theta_4 + Q \cos\theta_4 + R = 0$$

$\Downarrow$

$$P \frac{2 \tan\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)} + Q \frac{1 - \tan^2\left(\frac{\theta_4}{2}\right)}{1 + \tan^2\left(\frac{\theta_4}{2}\right)} + R = 0$$

$\Downarrow$

Position Analysis of Inverted Crank-Slider: Vector Loop (Cont.)

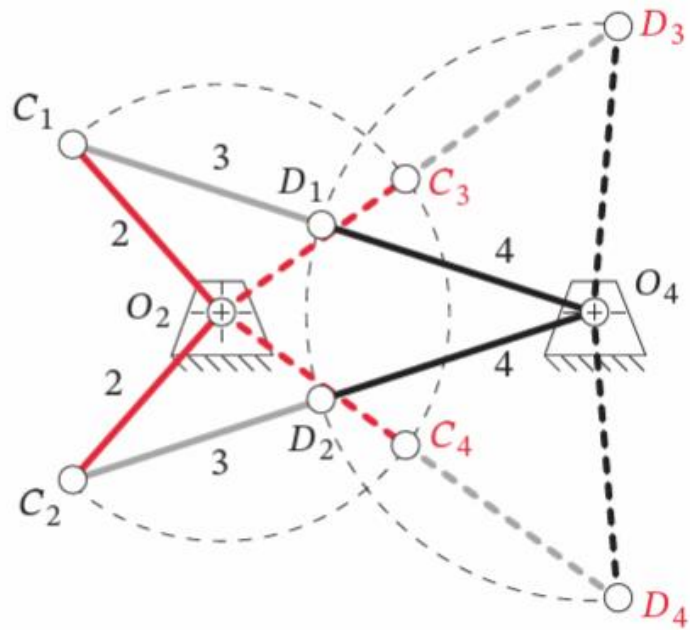
Find θ_3 from

open configuration: $\theta_3 = \theta_4 + \gamma$; crossed configuration: $\theta_3 = \theta_4 + \gamma - \pi$

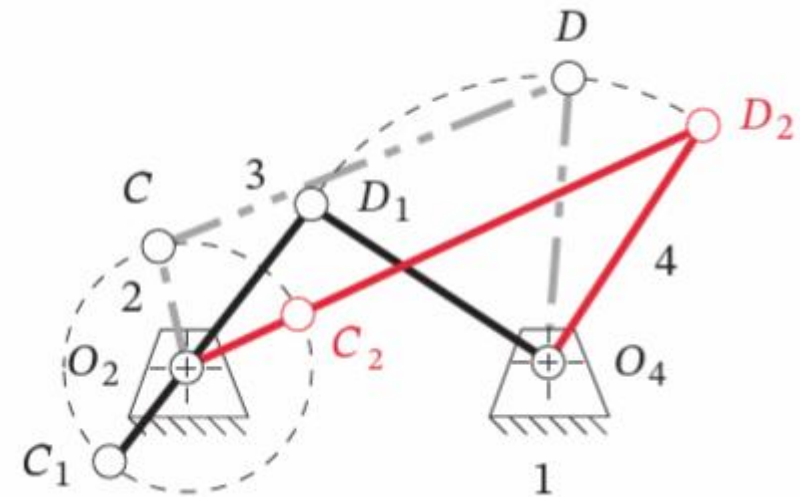
$$b = \frac{a \sin \theta_2 - c \sin \theta_4}{\sin \theta_3}$$

Motion Limiting Conditions

- **Collinearity** of two moving links
- **Toggle Position (Non-Grashof)**
- **Stationary Position (Grashof)**: Collinearity between crank and coupler



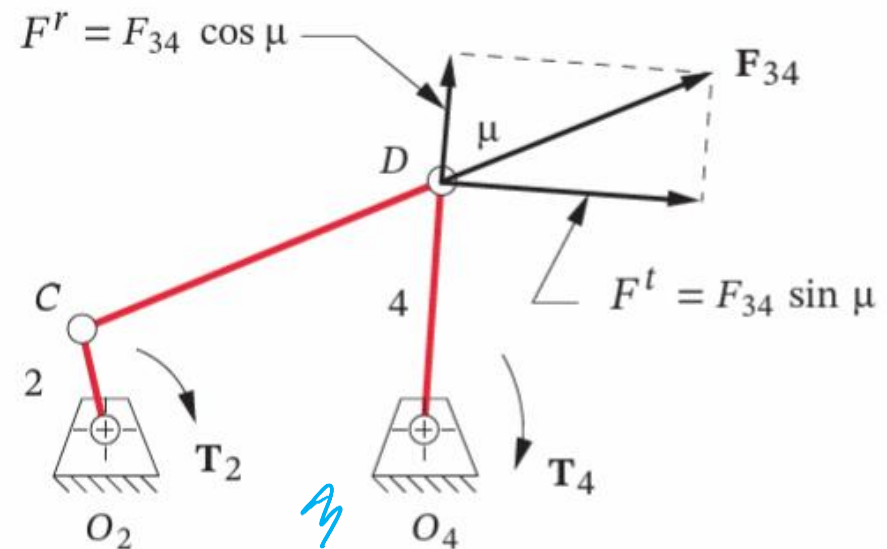
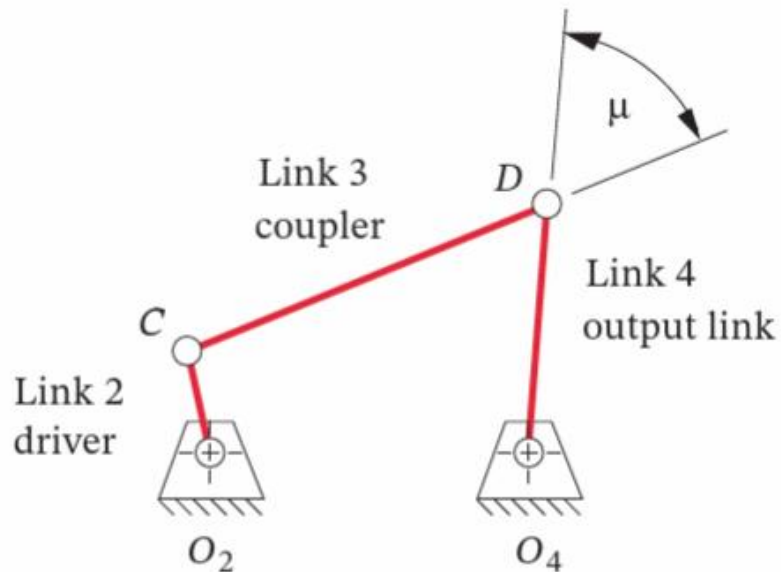
(a) Non-Grashof triple-rocker toggle positions



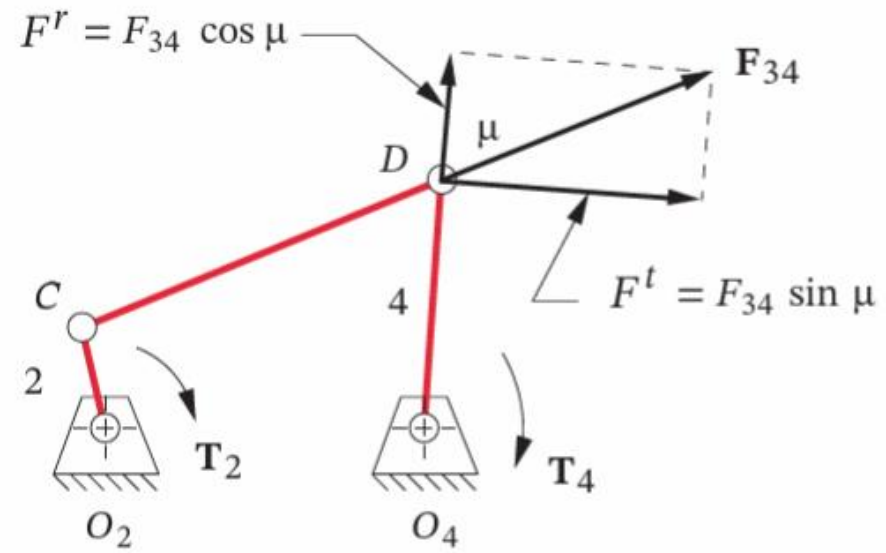
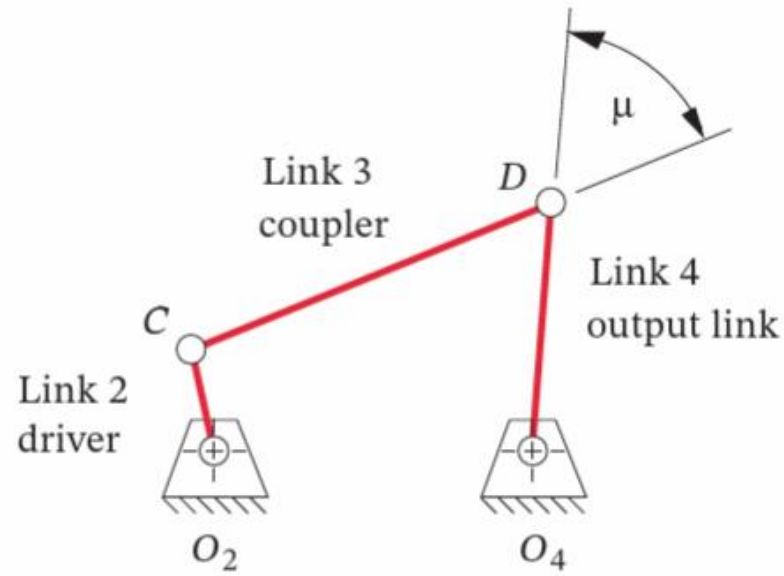
(b) Grashof crank-rocker stationary configurations

Transmission Angle

- The transmission angle is the angle between the output link and the coupler.
- For effective force transmission, the optimal transmission angle is 90 degrees; minimum angle > 40 degrees.



Transmission Angle (Cont.)



$$\theta_{\text{trans}} = |\theta_4 - \theta_3|$$

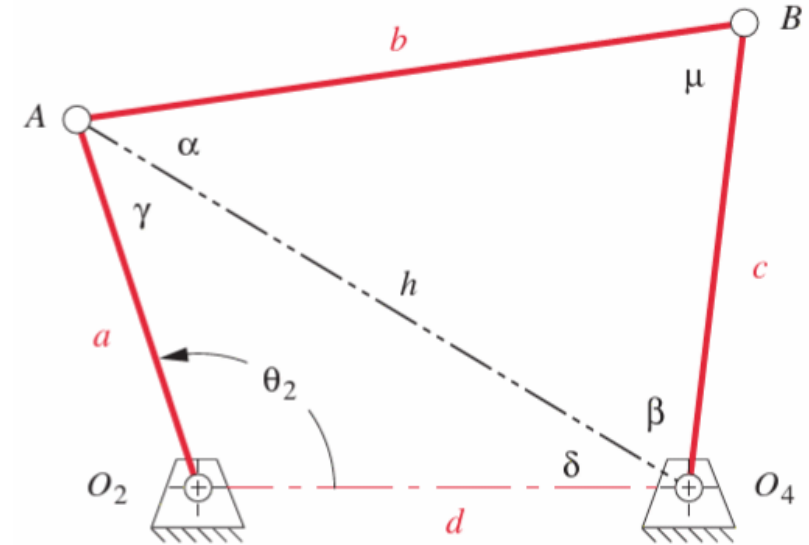
$$\mu = \begin{cases} \pi - \theta_{\text{trans}} & \text{if } \theta_{\text{trans}} > \frac{\pi}{2} \\ \theta_{\text{trans}} & \text{if } \theta_{\text{trans}} < \frac{\pi}{2} \end{cases}$$

Transmission Angle (Cont.)

Using Law of Cosines

$$h^2 = a^2 + d^2 - 2ad\cos\theta_2 = b^2 + c^2 - 2bcc\cos\mu$$

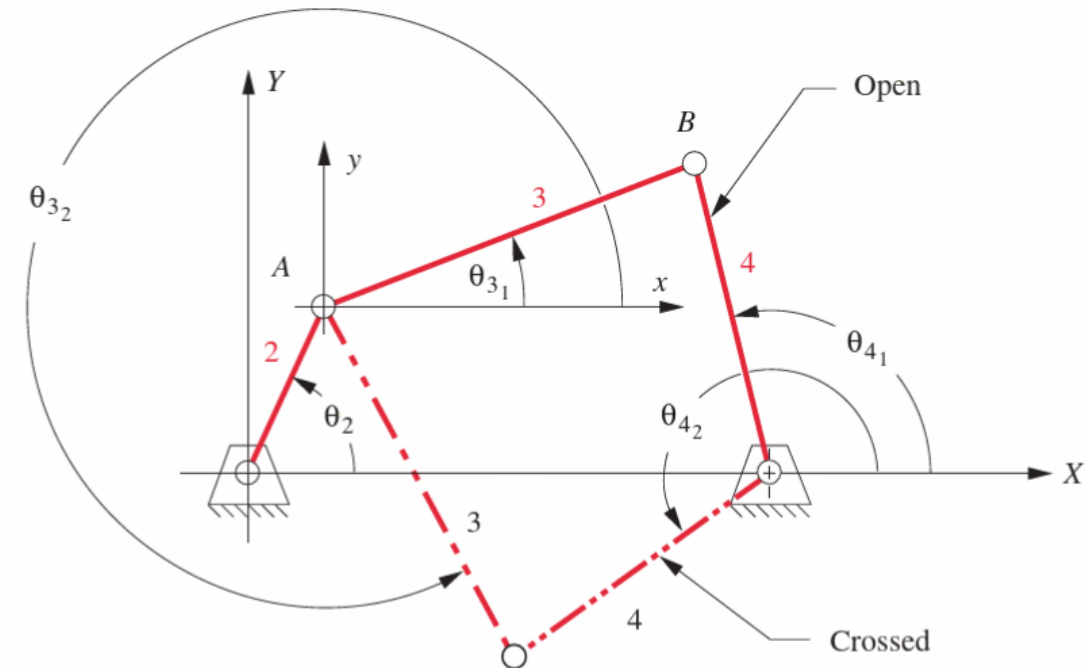
$$\cos\mu = \frac{b^2 + c^2 - a^2 - d^2}{2bc} + \frac{ad}{bc}\cos\theta_2$$



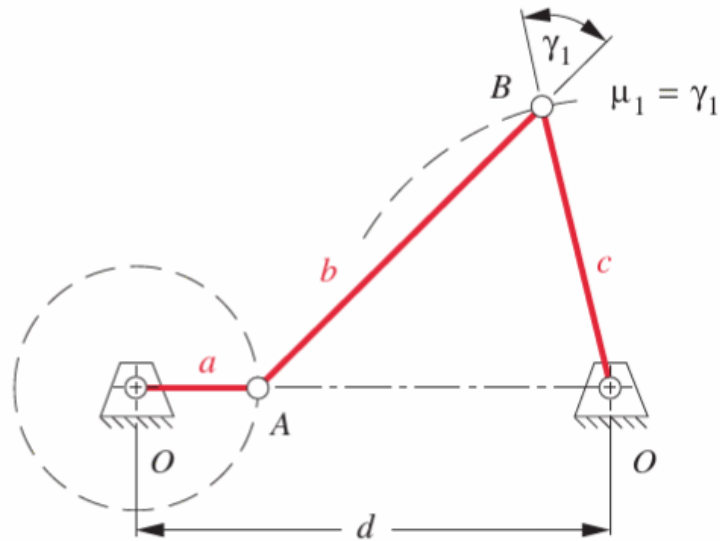
Transmission Angle: Example

*†4-13 Find the transmission angles of the linkages in the assigned rows in Table P4-1.

Link 1	Link 2	Link 3	Link 4	θ_2
6	2	7	9	30

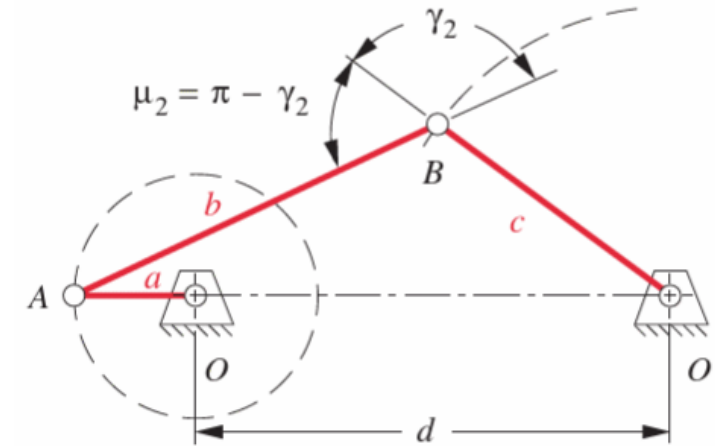


Extreme Transmission Angle: Grashof Crank-Rocker



(a) Overlapped

$$\mu_1 = \gamma_1 = \arccos \left(\frac{b^2 + c^2 - (d - a)^2}{2bc} \right)$$

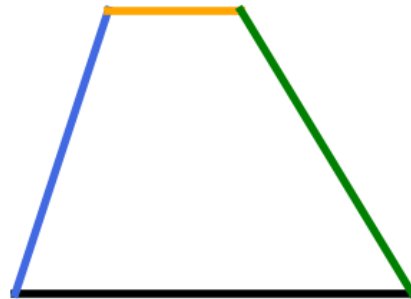


(b) Extended

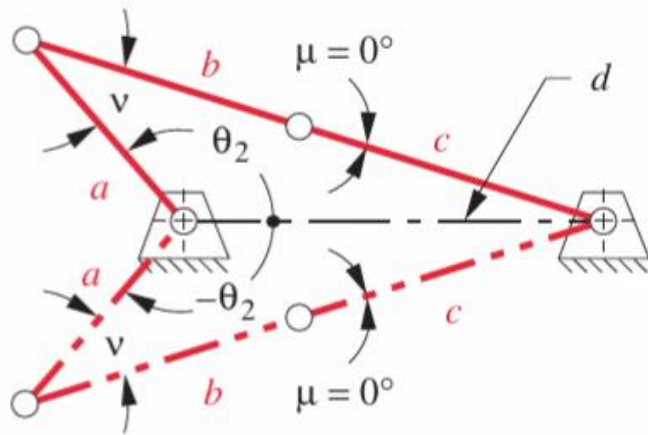
$$\begin{aligned} \mu_2 &= \pi - \gamma_2 \\ &= \pi - \arccos \left(\frac{b^2 + c^2 - (d + a)^2}{2bc} \right) \end{aligned}$$

Extreme Transmission Angle: Grashof Double-Rocker

- The transmission angle can vary from 0 to 90 degrees as the coupler can make full revolution with respect to other links.

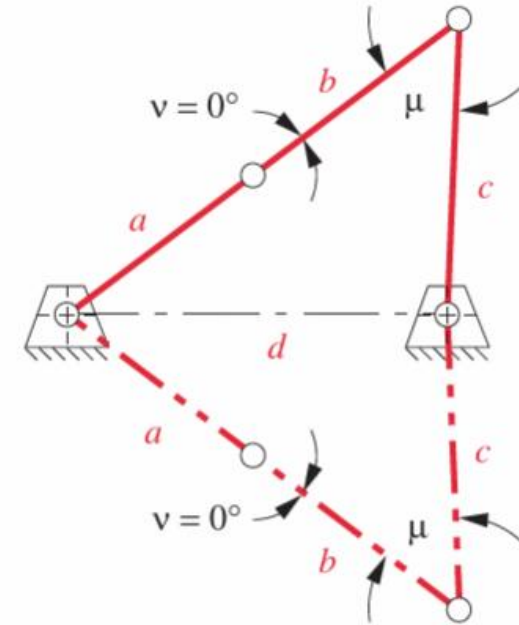


Extreme Transmission Angle: Non-Grashof Triple-Rocker



(a) Toggle positions for links b and c

$$\mu = 0$$

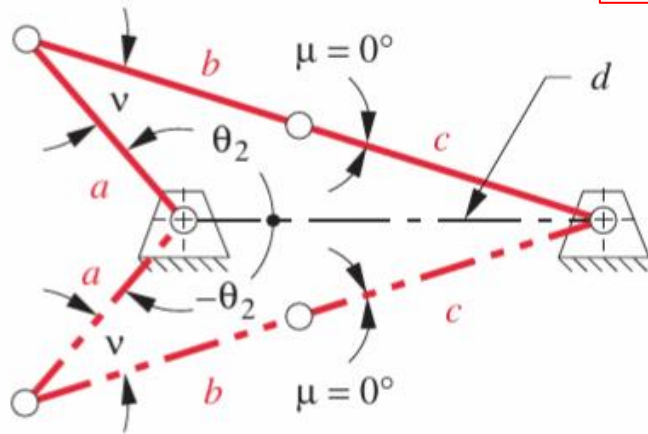


(b) Toggle positions for links a and b

$$\mu = \arccos \left(\frac{(a + b)^2 + c^2 - d^2}{2c(a + b)} \right)$$

Input Angle for Zero Transmission Angle

$$\mu = 0$$



(a) Toggle positions for links b and c

$$\cos\mu = \frac{b^2 + c^2 - a^2 - d^2}{2bc} + \frac{ad}{bc} \cos\theta_2$$

$$\pm 1 = \frac{b^2 + c^2 - a^2 - d^2}{2bc} + \frac{ad}{bc} \cos\theta_2$$

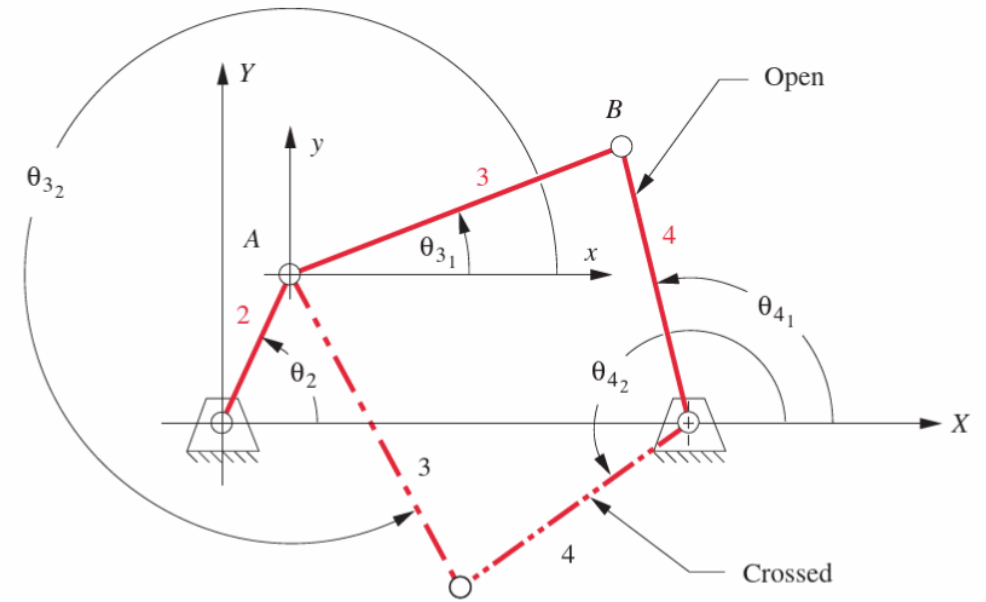
$$\cos\theta_2 = \frac{a^2 + d^2 - b^2 - c^2}{2ad} \pm \frac{bc}{ad}$$

$$\theta_{2_{toggle}} = \arccos\left(\frac{a^2 + d^2 - b^2 - c^2}{2ad} \pm \frac{bc}{ad}\right); \quad 0 \leq \theta_{2_{toggle}} \leq \pi$$

Input Angle for Zero Transmission Angle: Example

*†4-15 Find the input angles corresponding to the toggle positions of the non-Grashof linkages in Table P4-1. (For this problem, ignore the values of θ_2 given in the table.)

Link 1	Link 2	Link 3	Link 4	θ_2
20	10	10	10	50



Fourbar Quick-Return

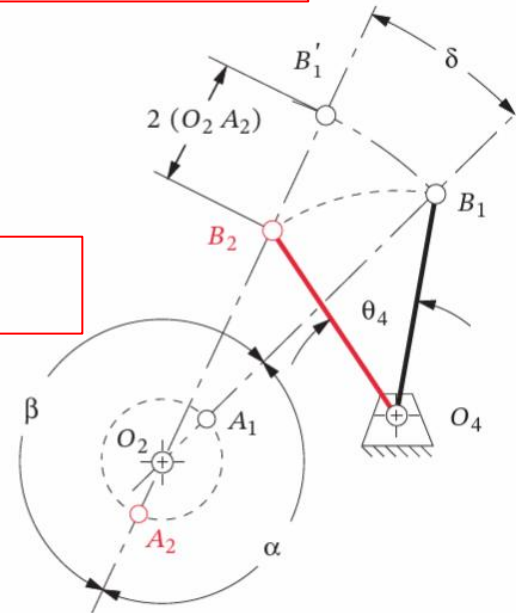
- _____ return: Equal time for the forward and the return stroke.
- _____ return: Return stroke accomplishes as quickly as possible.

Construction Angle: $\delta = |180 - \alpha| = |180 - \beta|$

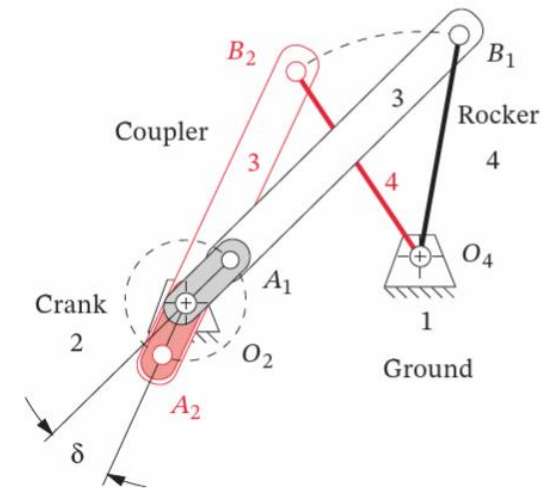
$$\alpha + \beta = 360$$

Time Ratio:

$$T_R = \frac{\alpha}{\beta} = \frac{\alpha}{360 - \alpha}$$



(a) Construction of a quick-return Grashof crank-rocker



(b) The finished linkage in its two toggle positions